

Preliminary Design and Performance Test of Tendon-Driven Origami-Inspired Soft Peristaltic Robot

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Abstract—Origami-based flexible, compliant and bio-inspired robots are believed to permit a range of medical applications with unpredictable environments. Here in this article, we experimentally demonstrate a novel origami inspired mobile robot structure which reconstructs its shape in the pivotal direction and launches peristaltic motion. To be able to sustain shear stress and counteract buckling, while accommodating the whole Origami robot in some stochastic confined environment, mobile robot sufficiently needed structural rigidity. To meet this requirement, Daler - Rowney canford paper of 150 gsm paper was chosen for carving fold lines. The Yoshimura origami pattern was manifested as the main body of the mobile robot to generate active deformation. In order to prevent buckling of springs during compression and expansion, special supporting structures made up of cardboard having alternate mountain and valley folds and coated in photopolymer resin. Finally, we were able to achieve a peristaltic motion of an average speed of 0.056 cm/sec, while we envision in future to incorporate the same construct for more potential biomedical and industrial application areas.

Keywords—Origami robot; flexible springs; friction coatings; soft robot.

I. INTRODUCTION

A. Basic Overview

Nature has always been an ideal source of inspiration for engineers to develop more robust, compliant and adaptable machines for effective and easy operation in complex natural environments [1]–[4]. Soft, flexible yet resilient, active deformation is one of the prime features envisioned for any soft robot [5]–[8]. Such robots have incorporated into the smart materials and soft technologies with the help of rapid prototyping tools to enhance the bio-inspired design [9]–[11]. For example, soft elastomeric materials can be designed with

biomaterials which are capable of delivering drugs locally [12] and further deposit some materials which can be used as a scaffold for tissue engineering [13]–[17].

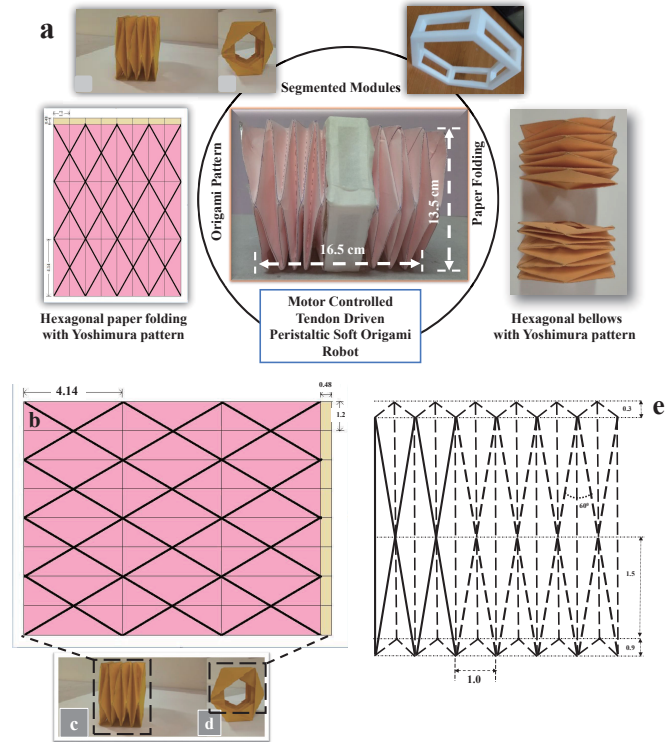


Fig. 1. (a) Prototype of the origami-inspired friction coated single motor controlled tendon-driven peristaltic soft robot. (b) Crease pattern for hexagonal bellows with Yoshimura origami design. The thin lines indicate valley folds and thick lines indicate the mountain folds. The yellow colored part demonstrated here is devoid of any crease pattern and has been added as an extra part to be rolled over and glued to another side of the paper to form a cylinder. In order to ease the process of making the bellows, the crease pattern made initially on the paper is unfolded (ensuring that the creases remain embossed) and then one shorter edge is rolled over to another shorter edge to form a cylindrical shape and glued. Then the paper is again folded according to the embossed pattern to form the final bellows. All dimensions are in cm (not to scale). One half of the hexagonal bellows structure with Yoshimura pattern in (c) front view and (d) side view. (e) Representation of how (a) will look at the end with final folding.

B. Origami Mechanisms

To actuate optimally with precise folding and to move in the stochastic environment, robotic engineers are recently looking upon century-old popular art of folding paper, “Origami”. Origami-based flexible, compliant and bio-inspired robots are

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believed to permit a large range of medical applications with unpredictable environments. Origami engineering demonstrates an important advantage to use 2D plane materials for 3D end applications.

C. Objective of the Present Study

We illustrate here in this article a proof-of-concept (PoC) analysis, the functional application of a crawling worm-like peristaltic locomotion mechanism with an initial prototype of service-robot utilizing monolithic, multi-faceted design for realizing crawling movement with a single servo motor. Foldable pattern oscillation is initiated by using this servo motor as a principle actuator. Daler - Rowney Canford paper of 150 gsm has been used creatively by folding it into the shape of bellows to provide enough flexibility and bendable rigidity. Due to its ability to change shape, we envision this origami robot to be used in future for applications where the robots are required to pass through narrow gaps and other such spaces which are difficult to navigate in a fully expanded state.

Conclusively, the origami-inspired Yoshimura pattern design based initial service robot prototype is proposed as a tendon-driven single motor controlled actuation mechanism in contrary to a sole passive structural element, presenting the structural realization for monolithic design. In the following sections, analysis related to the choice of best design for the shell as well as the servo motor and spring mechanism is illustrated. Subsequently, analysis and testing for the integrated module comprising both shell and actuator are shown in the context of the performance achieved, describing several characteristics extracted by rigorous experimental analysis. Finally, a conclusion with future work and discussion of the design is provided.

II. STRUCTURE OF THE MAIN BODY: YOSHIMURA ORIGAMI PATTERN

Amongst various types of popular origami folding methods (Miura-ori fold pattern, sail-folding pattern, water bomb pattern, Huffman grid), the Yoshimura pattern [18] was found to be optimal for our application since deformation was required horizontally [19]. Choice of appropriate paper for the hexagonal bellow structure was done by taking into consideration two main factors, namely flexibility and strength. Accordingly, Daler- Rowney Canford paper of 150 gsm, was found to be the perfect fit. Crease pattern for the hexagonal bellow structure was embossed in the paper after which, it was carefully folded into a combination of mountain and valley folds along this pattern. In this way, we built two hexagons (one outside and one inside) (figure reference needed) with an offset of 120 degrees from each other to accommodate in the final structure.

In this work, we have manually folded the whole design structure to form a chassis for our mobile robot, while we envision to use commercial rapid prototyping tool (drawing software tool such as ORIPA, Rigid Origami Simulator, TreeMaker, Freeform Origami etc.) in near times. The significant parameters and to creatively build a 3D model from a 2D structure is a captivating multidimensional aspect, a bit of

which is depicted here in Fig. 1. The length and width of each cell were chosen (Fig. 1) so as to ensure there was sufficient space to house the servo motor inside the hexagonal bellows after they are formed, while the dimension of each and every folding is been addressed optimally inside Fig. 1.

III. DESIGN AND FABRICATION OF THE TENDON-DRIVEN ORIGAMI ROBOT

The main body of the origami robot consists of two hexagonal bellow structures linked together by a 3D printed motor holder with a servo motor (Fig. 2). Each of the bellow structure created out of Daler-Rowney Canford paper (150 gsm) houses a spring module made up of two submodules each and supported by hexagonal 3D printed plates. A nylon thread running through the center of these plates *via* thread holders and springs attaches to the extended shaft of the servo motor and gets wound in specific direction upon the rotation of the motor. Cardboard structures folded using alternate mountain and valley folds and coated with photopolymer resin have been used to provide strength and prevent buckling of springs.

A. Design of Sub-modules

Open-coil helical stainless-steel compression springs have been used in the design of each submodule. Each spring is placed between two hexagonal plates and two such springs in series constitute one module which is housed inside one hexagonal bellow. Similar construction pattern is followed for the other hexagonal bellow. In order to prevent buckling of springs during compression and expansion, special supporting structures made up of cardboard having alternate mountain and valley folds and coated in photopolymer resin have been included. The parameters of the chosen spring are depicted in Table I below.

TABLE I
SPRING PARAMETERS OF THE ORIGAMI-BASED DEFORMABLE ROBOT

Parameters	Value
Solid height	4.12 mm
Spring constant	0.006 kg/mm
Wire diameter	0.5 mm
Free length	40 mm
Mean diameter	11.5 mm

B. Selection of Spring

The correct choice of springs was made after taking into consideration the following points.

1) *Solid Height*: The solid height of a compression spring is defined as the height of the spring when all of the coils are brought together (stacked) by compression (load). The solid height of the spring was equal to the length of the origami structure when it was compressed to the maximum extent, keeping a safe margin. The solid height of the spring chosen was 4.12 mm.

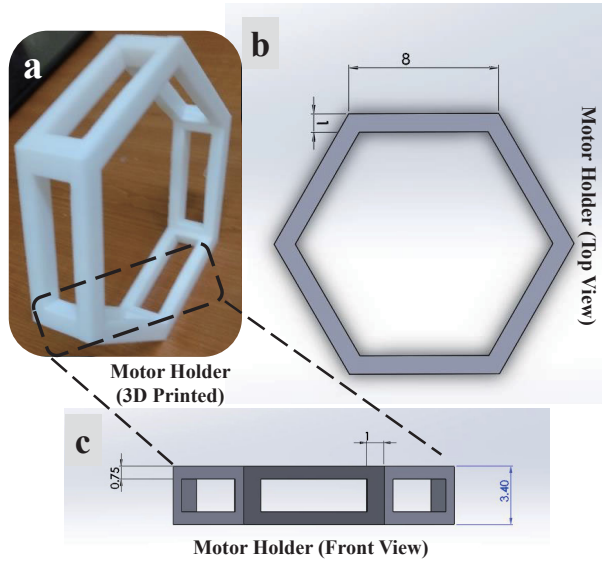


Fig. 2. (a) PLA based 3D printed motor holder. Motor holder (b) in Top view (c) and front view. Motor is fixed on the bottom hexagonal side with reference to (b). The motor holder is used to link the two hexagonal bodies on its both sides. Dimensions are in cm (not to scale).

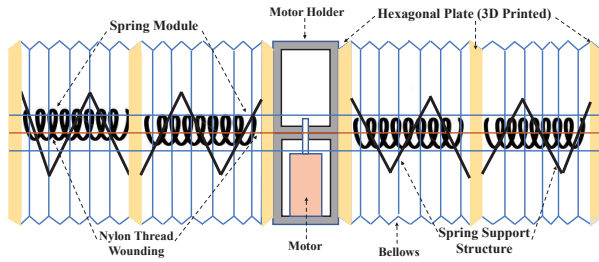


Fig. 3. Overall internal structure of the robot. 3D printed module for deformation mechanism consisting two submodules in series both sides of the motor. The motor is fastened to one side of the motor holder while the wires of the motor come out from the motor holder structure due to the gaps left in the structure. The wires are further connected to the Arduino board. Each module is located on each side of motor holder.

2) *Spring Constant*: Spring constant is an imperial characteristic defined as the ratio of force affecting the spring to the displacement caused by it. The spring constant was chosen optimally to be as less as possible so as to facilitate easy compression of the spring and also to exert the least amount of torque on the servo motor. The spring chosen had optimized dimensions having a minimum spring constant of 0.006 kg/mm.

3) *Wire Diameter*: It is the diameter of the coil of the spring. The wire diameter of the spring was chosen 0.5 mm so as to accommodate inside hexagonal plate properly.

4) *Free Length*: The actual length of a compression spring without any sort of load or force being placed on it is called the free length of the spring. The free length of the spring must be equal to the length of the origami structure in its expanded

state. The free length of the chosen spring was 40mm for desired mechanical stiffness.

5) *Mean Diameter*: It is the diameter between the inner and outer diameter of the spring. The mean diameter of the chosen spring was 11.5 mm.

C. Spring Buckling & Slenderness Ratio

Buckling of spring is a phenomenon in compression springs which can cause undesirable bending and motion in a non-axial direction. Hence it is necessary to keep buckling within the limit. This can be done by maintaining the coil spring slenderness ratio below 4. The slenderness ratio presented here was 3.2. To give strength to the springs and also as a precaution against slightest buckling that may occur due to the weight of the hexagonal plates, the springs were housed in a cardboard structure designed using mountain and valley folds. Individual cardboard square of dimension 4cm x 4cm was cut and joined using paper tape to add strength at the joints. Additionally, the structure was coated with photopolymer resin and cured in XYZ printing UV curing chamber to give extra stiffness and to make it robust. Appropriate size hole was drilled through it to facilitate hindrance-free movement of spring while operation.

D. Selection of Motor

The actuator was selected considering various factors such as torque, speed, size, and weight. The size and weight were required to be at least as possible to accommodate the motor inside the motor holder and not exert too much force on it. The torque must be enough to overcome friction between bellows and ground while the speed should facilitate planar actuation without any buckling. Thus a single motor was used to actuate both sides simultaneously.

The standard servo can rotate approximately 180 degrees (90 degrees in each direction) while, continuous servo motor rotates fully forward or backwards instead of moving to a position at the given speed in that direction (Table II).

TABLE II
MOTOR SPECIFICATIONS OF THE TENDON DRIVEN ORIGAMI INSPIRED SOFT PERISTALTIC ROBOT

Parameters	Value
Model	SM-S4315R large cont. rotation plastic Servo
Weight	0.09 kg
Operating speed	53.0 R/M (4.8V), 62.0 R/M (6V)
Stall torque	15.4 kg-cm, 213.9 oz-in (4.8V)
Stall torque	14.5 kg-cm, 201.4 oz-in (6V)
Rotation angle	360 degree

IV. DEFORMATION MECHANISM: ACTUATION

The compression springs attached in series allow the robot to expand and compress in the horizontal direction. The nylon thread (diameter 0.4mm) is attached to the servo in a manner such that whenever servo rotates, the thread from one bellow structure gets wound, while that from another bellow structure is unwound, thus compressing and expanding respective sides (Fig.3). The high stiffness and strength of nylon thread assist

in the easy unwinding of thread from the motor shaft. The entire robot is actuated using a single continuous rotation (360 degrees) servo motor. The torque exerted by the motor is sufficient to overcome ground resistance due to friction while the speed (rpm) is appropriate to just provide planar motion without any buckling.

V. MOTION PLANNING & CONTROL OF THE ROBOT

The robot was actuated by continuous rotation servo which was programmed and controlled using Arduino Uno board. Two types of motions have been demonstrated, namely, simultaneous (expansion/compression) motion and peristaltic motion. In the simultaneous motion, both the sides of the robot compressed and expanded at the same time depending upon the clockwise or counterclockwise rotation of the motor. Whereas, in the peristaltic motion, while one side of the robot compressed, the other side expanded and *vice-versa*. In order to initiate the peristaltic motion, the initial positioning of the robot was made such that one side remain compressed while the other side remained expanded. Thus, a peristaltic motion was achieved by properly adjusting the time duration, direction, and speed of servo motor. As a result, the robot received a slight pull in the forward direction when that part expanded while the other part compressed. The friction of the rear part was slightly increased so as to prevent unnecessary backward motion.

A. Calculating Velocity of Peristaltic Locomotion

Servo motor installed inside the origami inspired robotic main body transfers rotational energy to the spring module as elastic energy. Here for the sake of simplicity, we assume, there is negligible energy lost during transformation into spring elastic motion generation. So the resultant force responsible for peristaltic motion generation is described as follows.

Here, we consider $F_{Resultant}$ as the resultant force acting on the origami inspired robot main body as a function of imparted force in the spring modules. $F_{Applied}$ is the force generated from servo motor module and $F_{Friction}$ is the counteracting force due to surface irregularities.

$$F_{Resultant} = F_{Applied} - F_{Friction} \quad (1)$$

The spring module is assumed to have an effective spring constant as k_{Eff} while μ_{Static} is considered to be the static friction co-efficient between origami service robot and the surface.

$$F_{Resultant} = k_{Eff}X^*(L, t) - \mu_{Static}N \quad (2)$$

Substituting $X^*(L, t)$ and incorporating velocity $V^*(L, t)$ in deformed state of a point on the origami main body at the (un-deformed) position L and time t ,

$$m \frac{dV^*(L, t)}{dt} = k_{Eff} \int_0^L \varepsilon(l, t) dl - \mu_{Static} m \cdot g$$

$$i.e. \quad V^*(L, t) = \frac{k_{Eff} \int_0^L \int_0^t \varepsilon(l, t) dl dt}{m} - \mu_{Static} \cdot g \cdot t$$

Here m denotes the mass of the whole robotic body while g is the acceleration due to gravity. This is an analytical approach through which we can calculate the position as well as velocity of the robot peristaltic locomotion which will be further discussed with experimental demonstration.

VI. EXPERIMENTAL OBSERVATIONS

A. Material Property : Mechanical Test

To find out plastic flow behavior, strength, and ductility, mechanical stress/strain analysis were performed in INSTRON UTM (universal testing machine) with and without spring module (Fig. 4). Our origami inspired service mobile robot weight also optimized for slow but steady initial axial motion translation (Table III).

TABLE III
TENDON DRIVEN ORIGAMI INSPIRED SOFT PERISTALTIC ROBOT SPECIFICATIONS

Parameters	Value
Total robot weight	407.20 gm
Weight of one half origami structure	21.09 gm
Weight of motor holder (joint)	59 gm
Weight of hexagonal plate	23 gm
Weight of one thread holder	1 gm
Length of one hexagonal bellow	8.5 cm
Width of motor holder	3.5cm
Total length in relaxed position	20.5 cm
Total length in compressed position	9.5 cm

Rigid yet extremely lightweight polystyrene sheet was cut and stick with INSTRON machine to introduce equilateral tensile compression during a test. We repeated each experiment at least three times for data reproducibility and expressed here as mean $\pm$ standard error unless otherwise specified. To illustrate elasticity we calculate Young's modulus of each module separately.

1) *Theoretical stiffness evaluation:* The spring constant of the chosen spring was 0.006 kg/mm as specified earlier. So, as they are in parallel configuration the combined spring constant is 0.012 (= 2*0.006) kg/mm. As from the experimental observation we managed to calculate Young's modulus value of each module and hence we would like to validate the same with our design parameters.

2) *Origami module Young's modulus evaluation:* Tensile compression test as performed with several trials, we are able to quantify mean Young's modulus of each origami module. Mean Young's modulus of each origami module with spring was 330.40 Pa while without spring module i.e. with only Canford paper came out to be 275.30 Pa (Fig. 4 a). Based on Young's modulus defined here, an internal material property that demonstrates an early stage parameter estimation, mimics closely how the deformation characteristics will be. We are here approximating the structure to be inside elastic domain (Neo-Hookean material), continuum and hence viscoelasticity is been neglected for primary consideration. These values justify deflection characteristic and paper thickness as required to meet optimal axial motion behavior.

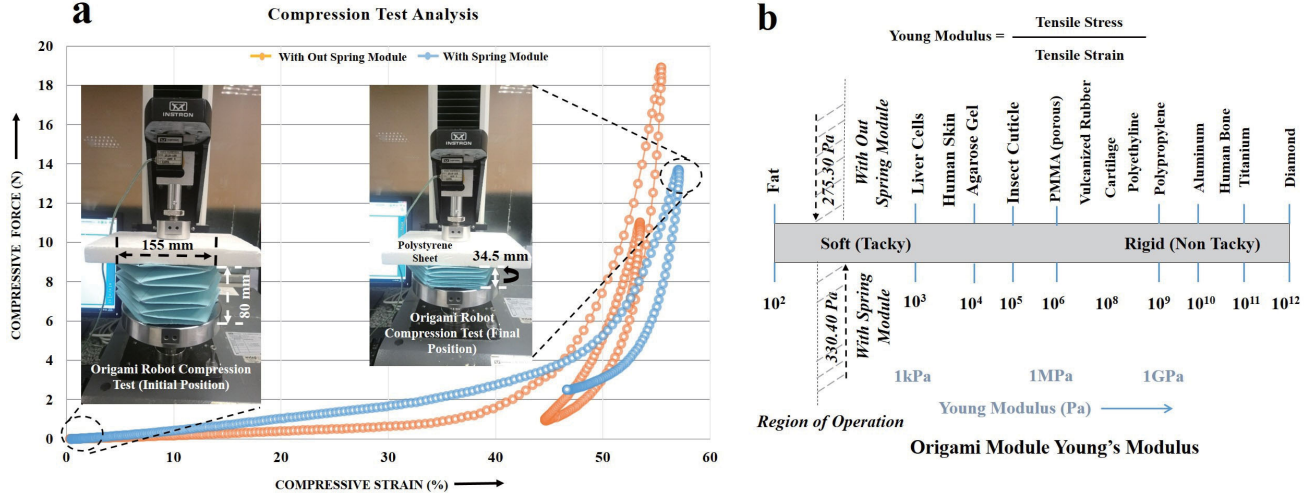


Fig. 4. Compression test with and without spring module in INSTRON UTM. (a) Inset dimension for pre and post compression testing (not to scale). For symmetric compressive force transmission on origami based robot surface, polystyrene sheet was used. (b) The transition from soft to rigid materials by Young's modulus [9]. The shaded region is the region of operation for Origami inspired spring module.

B. Friction Quantification

To determine surface roughness and quantify friction between the robot and contact medium, we performed trials with a force sensor and a test weight of 2 kg as shown in (Fig. 5). The coefficient of static friction μ_{Static} between the surface of mobile robot (Canford paper) and different surfaces e.g white board and cylindrical pipe made of PET is further calculated to quantify and demonstrate friction coating mechanism effectively. Here, we would like to mention that we have optimally characterized the friction as quantified here under while less friction will enhance sliding and increasing friction will hinder the speed of locomotion. So, this is a trade-off problem to be solved while designing the friction pads likewise. Here, in static condition, a force sensor is being introduced to calibrate static friction and recorded just prior to motion as demonstrated in Fig. 5.

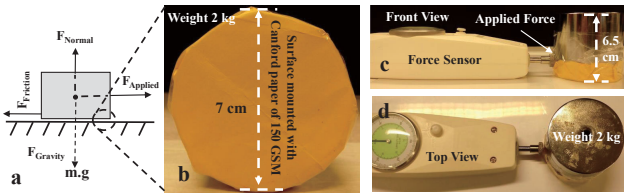


Fig. 5. Friction quantization using a sample 2 kg weight. (a) Free body diagram (b) Bottom surface of 7 cm diameter cylindrical 2 kg sample weight embedded with Canford paper to test the friction coefficient. (c) & (d) depicts a force sensor and the sample for friction measurements.

To quantify friction in the different surfaces we define $F_{Applied}$ as applied force sensed from force sensor; $F_{Friction}$ as restoring force due to friction; F_{Normal} as the normal force

of the sample weight; and $F_{Gravity}$ as the force due to gravity. So,

$$F_{Gravity} = 2 \times 9.81 = 19.62 \text{ kg.m.s}^{-2}. \quad (3)$$

During balanced condition,

$$F_{Friction} = \mu_{Static} \times F_{Normal} \quad (4)$$

$$F_{Normal} = F_{Gravity} \quad (5)$$

$$F_{Friction} = F_{Applied}. \quad (6)$$

As per force sensor reading the time when sample 2 kg weight is about to be in motion,

for White Board :

$$F_{Applied} = 7N$$

for PET Sheet :

$$F_{Applied} = 5N$$

$$\left\{ \begin{array}{l} \text{for White Board: } F_{Applied} = 7 \text{ N} \\ \text{for PET Sheet: } F_{Applied} = 5 \text{ N} \end{array} \right\}$$

i.e.

$$\left\{ \begin{array}{l} \mu_{Static_{Whiteboard}} = 7/19.62 = 0.356 \\ \mu_{Static_{PET}} = 5/19.62 = 0.254 \end{array} \right\}$$

C. Inspection & Surveillance : An Initial Attempt

In this article we intend to demonstrate origami inspired service mobile robots for peristaltic motion inside confined areas. The bi-directional origami robot as illustrated here can change its shape either by compressing and expanding both the sides simultaneously or alternately giving rise to peristaltic motion (Fig. 6). Initially origami pattern tried and tested for making the structure of the main body. The origami robot deforms in horizontal as well as vertical direction. The

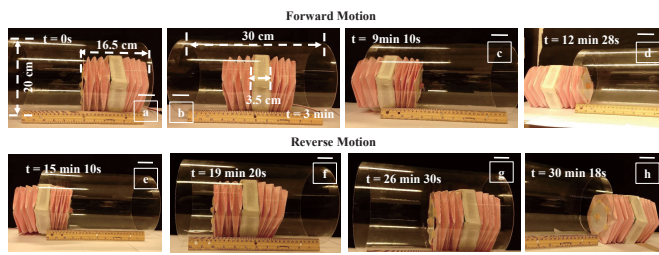


Fig. 6. Time lapse images for bi-direction peristaltic motion of origami inspired mobile soft robot inside a pipe of 20 cm diameter. Time duration of 12 min 28s were taken for axial movement to traverse the length of pipe i.e. 30 cm in forward direction (a-d). Time duration of 15 min 8 sec were taken for the in reverse direction (e-h). Motor module is housed inside soft robot with 3.5 cm thickness as demonstrated in (b). Scale bar 5 cm.

deformation in horizontal direction is desired since it helps in compressing and expanding the structure along the horizontal axis. Here, on a white board we are able to achieve an average speed of peristalsis as 0.056 cm/sec while on PET sheet the speed was 0.040 cm/sec. The same service robot takes 15 min 8 sec to return back in PET sheet with an average speed of 0.033 cm/sec (Fig. 6).

Here, while changing directions we swapped the friction coatings in our bi-directional friction coated mobile origami inspired service robot for reverse traversal. The movement of the robot can be made faster by using higher RPM servo and the control can be made fully autonomous by making necessary changes in the code. Due to initial prototype the motion is slow and the control methods are rudimentary. This initial inspection and monitoring task is very much in line with the need to open clogging, smooth transportation of drinkable water, effluent water, fuel oils, and gas. The same prototype has also been used on different surfaces to demonstrate multinational and diversified applications. We can extrapolate the same design and downgrade dimension in future to use in surgery and medical robotics *in vivo*.

VII. CONCLUSION & FUTURE WORK

Flexible robots can be sent inside narrow and confined spaces, while it sent through a pipe or tube, the flexible robotics and navigation play an important role. In the heavily populated areas such as at Singapore, there are wide varieties of pipelines e.g. gas lines, sewage, drainage system, water transportation etc. to run the city efficiently. In this context, among others, clogging is a serious issue faced in Singapore community for which inspection and navigation robots with shape adaptation is a must choice. In this paradigm, our origami based flexible, compliant and bio-inspired robots are believed to serve applications in unpredictable environments with flexible shape transformation capabilities. The soft flexible robotic system with embodied navigation intelligence can bypass obstacles and adapt its shape to be compliant with the curvy passage, and have direct access to the target sites under proper planning and navigation. Key challenging problems will be investigated in terms of design, characterization, planning,

tracking, navigation and validations for the new soft flexible robotic paradigm, which is expected to yield controllable, flexible, maneuverable, conformable and agile multiple-degree-of-freedom manipulations in an effective and compliant way.

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